

DexEyeGrasp: Learning Dexterous Grasping in Clutter with FingerEye Observations

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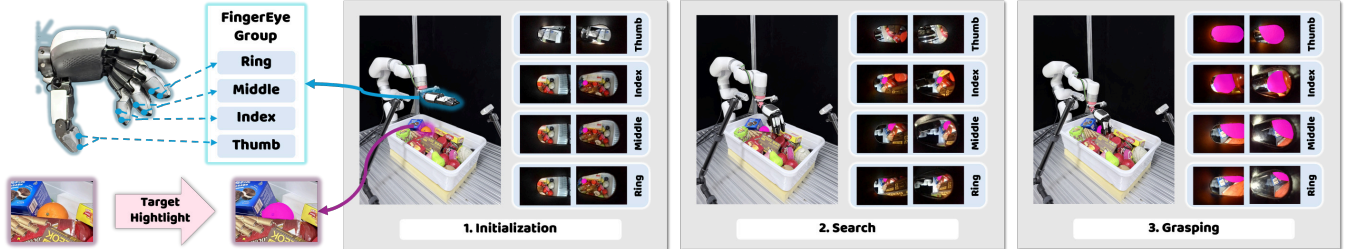


Fig. 1. An example of FingerEye-centric grasping in dense clutter. The FingerEye Group comprises all fingertip cameras on the hand, arranged as binocular RGB pairs. (1) Initialization: the designated target is highlighted by a magenta target-mask overlay in the FingerEye observations. (2) Search: the hand uses FingerEye observations to locate and approach the target in clutter. (3) Grasping: FingerEye observations guide finger and wrist motion to establish a grasp on the target.

Abstract—Dexterous grasping in dense clutter requires precise finger and wrist control under severe occlusion, which can hide the target and spatial cues needed for grasping from third-person cameras. We present DexEyeGrasp, a visuomotor learning framework that leverages FingerEye observations to guide both finger and wrist motion. Our framework distills a state-based teacher trained in simulation into a visuomotor student policy that integrates multiple dexterous viewpoints. To help the policy extract task-relevant spatial information, we introduce self-supervised learning objectives that predict current and action-conditioned future camera–target geometry. These objectives encourage geometric awareness for approaching the target and refining contact-rich interactions despite changing visibility. To further address the visual sim-to-real gap, we learn a residual image encoder from real FingerEye trajectory sequences while keeping the simulation-trained network frozen with aforementioned objectives. Together, these designs support learning and transferring dexterous grasping in densely cluttered environments.

I. INTRODUCTION

Dexterous grasping in clutter requires a robot hand to interact with multiple objects before establishing a stable grasp on the desired target. Unlike grasping on a clean tabletop, the target may be partially visible or even tightly surrounded. Successful manipulation therefore requires precise control of the hand amid surrounding objects [1], [2]. This problem presents two fundamental challenges. First, the high-dimensional actuation of dexterous hands, together with discontinuous and complex contact dynamics, makes it difficult to discover effective interaction strategies. Second, severe occlusions caused by neighboring objects and the robot hand make it difficult to perceive the target and its spatial relationship to the hand and surrounding objects.

A common approach in dexterous manipulation is to train a state-based teacher through reinforcement learning in simulation and then distill its behavior into a student policy using deployable observations [3], [4], [5], [6], [7], [8], [1],

[2]. With access to privileged simulation states, the teacher can accurately observe the target pose, its complete geometry, and the spatial relationships between the target, the hand, and surrounding objects. Such information supports learning contact-rich grasping behaviors. To deploy these behaviors using visual observations, existing approaches commonly employ target segmentation and tracking models, such as SAM2 [9], to help the student distinguish the desired object from clutter.

However, this paradigm introduces a fundamental mismatch between the information available to the teacher and that observed by the student. A third-person camera observes the scene from a fixed viewpoint, where occlusion can obscure the target and the spatial cues needed to guide wrist motion and finger placement [10]. Moreover, as the hand approaches the target, the most task-relevant regions are increasingly occluded by the hand itself. As a result, the performance of segmentation and tracking models can become unstable under such heavy occlusions [8]. Consequently, the visual student is required to reproduce contact-rich behaviors using weaker information than that available to the teacher.

FingerEye cameras [10] provide fine-grained observations from dexterous viewpoints enabled by articulated finger motion. The FingerEye Group comprises all fingertip cameras on the hand, arranged as binocular RGB pairs on the equipped fingertips (Fig. 1). Their images collectively form the FingerEye observations, whose viewpoints change with both finger articulation and wrist motion. These dexterous viewpoints can reveal targets occluded from other views and convey spatial relationships among the hand, target, and surroundings for wrist approach and precise finger placement. Nevertheless, learning from FingerEye observations is non-trivial. Their viewpoints change rapidly with hand motion, their fields of view only partially overlap, and the target may enter, leave, or reappear in individual views. The

resulting visual observations contain substantial variations caused by the hand, background, and non-target objects. Such dynamic and task-irrelevant variations can interfere with policy learning and further amplify the visual domain gap between simulation and the real world.

In this work, we introduce DexEyeGrasp, a FingerEye-centric visuomotor framework for dexterous grasping in clutter. Our framework learns directly from dexterous observations captured by multiple FingerEye cameras. We design an efficient visuomotor architecture that selectively extracts and integrates task-relevant information across these dexterous viewpoints. Furthermore, we introduce FingerEye-centric self-supervised objectives that encourage the visual representation to capture interaction-relevant geometric and motion cues, thereby improving both policy learning and sim-to-real transfer.

Our main contributions are:

- We propose a FingerEye-centric visuomotor architecture that integrates multiple dexterous viewpoints to capture spatial information for coordinated finger and wrist control during grasping in clutter.
- We introduce self-supervised learning objectives that suppress task-irrelevant visual variations and improve the sim-to-real transfer of FingerEye.
- We demonstrate strong performance across challenging simulated clutter configurations and successful transfer to real-world dexterous manipulation.

II. RELATED WORK

A. Dexterous Grasping in Clutter

Recent work demonstrates promising dexterous grasping in clutter. RetrDex [8] focuses on exposing buried targets rather than completing grasps. Adding a grasping module would introduce policy-switching challenges. DexSinGrasp [1] and ClutterDexGrasp [2] instead learn clutter interaction and grasping within a single policy. DexSinGrasp exploits finger dexterity for coordinated singulation and grasping, but its main formulation centers the target in a dense clutter. ClutterDexGrasp learns obstacle clearance and grasping through reward shaping and curricula, with visible targets from a fixed third-person camera. We instead study extremely dense box clutter with randomized target poses, where distractors, the robot, and box walls severely occlude third-person viewpoints. To support grasping in this setting, we integrate FingerEye observations from dexterous viewpoints for fine-grained perception.

B. Visuomotor Policy Learning

Recent advances in visuomotor policy learning enable robots to acquire complex manipulation skills directly from visual observations. Existing approaches typically learn policies from third-person or wrist-mounted cameras [11], [12], [13]. Occlusion can limit target visibility in these views and obscure the spatial relationships needed for precise manipulation. FingerEye [10] introduces fingertip visual feedback for fine-grained manipulation across object sizes and initial

poses with fixed wrist. We extend this sensing to dense-clutter grasping, where FingerEye observations guide both finger and wrist motion to the target object under occlusion. Our self-supervised objectives encourage the policy to extract target-relative geometric cues from FingerEye observations. Such geometric awareness is particularly valuable for precise grasping in cluttered environments.

C. Sim-to-Real Visual Transfer

Visual sim-to-real transfer remains a major challenge for visuomotor policies learned in simulation. Extensive visual randomization is commonly used to improve transfer [14], [15], [16], [17]. However, randomization in simulation may not adequately cover real-world visual distributions in complex scenes, motivating adaptation with real data. One option is to co-train a shared policy on simulated and real demonstrations [18], [19], [20]. Yet real action labels are scarce, and teleoperated behaviors can differ from those learned through reinforcement learning; naive co-training therefore risks disrupting the simulation-learned policy [21]. Additional visual adaptation [22] offers an alternative that retains the original network while learning feature corrections for real images. We train a residual image encoder on real FingerEye data, keeping the simulation-trained network frozen. Supervision through its current and next-step geometric prediction heads aligns real-image features with the image feature in simulation, aiming to reduce the visual domain gap without directly imitating teleoperated actions.

III. PROBLEM FORMULATION

A. Task Definition

We study dexterous grasping of a designated target among N objects, with the remaining objects referred to as distractors. The robot must grasp the target and transport it to a fixed goal $\mathbf{p}_g$ above the support surface. The support area A , wall height h , and object count N control clutter density and the space available for moving distractors. Success requires valid target-hand contact within a goal-distance threshold $\varepsilon_g = 0.10\text{m}$ for $H_g = 15$ consecutive policy steps.

We use teacher-student distillation: a state-based teacher trained with privileged information in simulation supervises a deployable student using only proprioceptive signals and visual observations.

B. State Space

We represent the teacher state as $\mathbf{s}_t = [\mathbf{x}_t^{\text{prop}}, \mathbf{x}_t^{\text{priv}}]$, where $\mathbf{x}_t^{\text{prop}}$ contains deployable proprioceptive signals and $\mathbf{x}_t^{\text{priv}}$ contains information available only in simulation. The proprioceptive state comprises hand joint positions and velocities, end-effector position and 6D orientation, end-effector linear and angular velocities, fingertip positions, and the previous action. The privileged state comprises the target pose and velocities; the target pose in the palm frame; fingertip offsets from the target center, together with distances and unit directions from the nearest target surface points to the fingertips; distractor positions relative to the target, linear velocities, displacement magnitudes from their initial

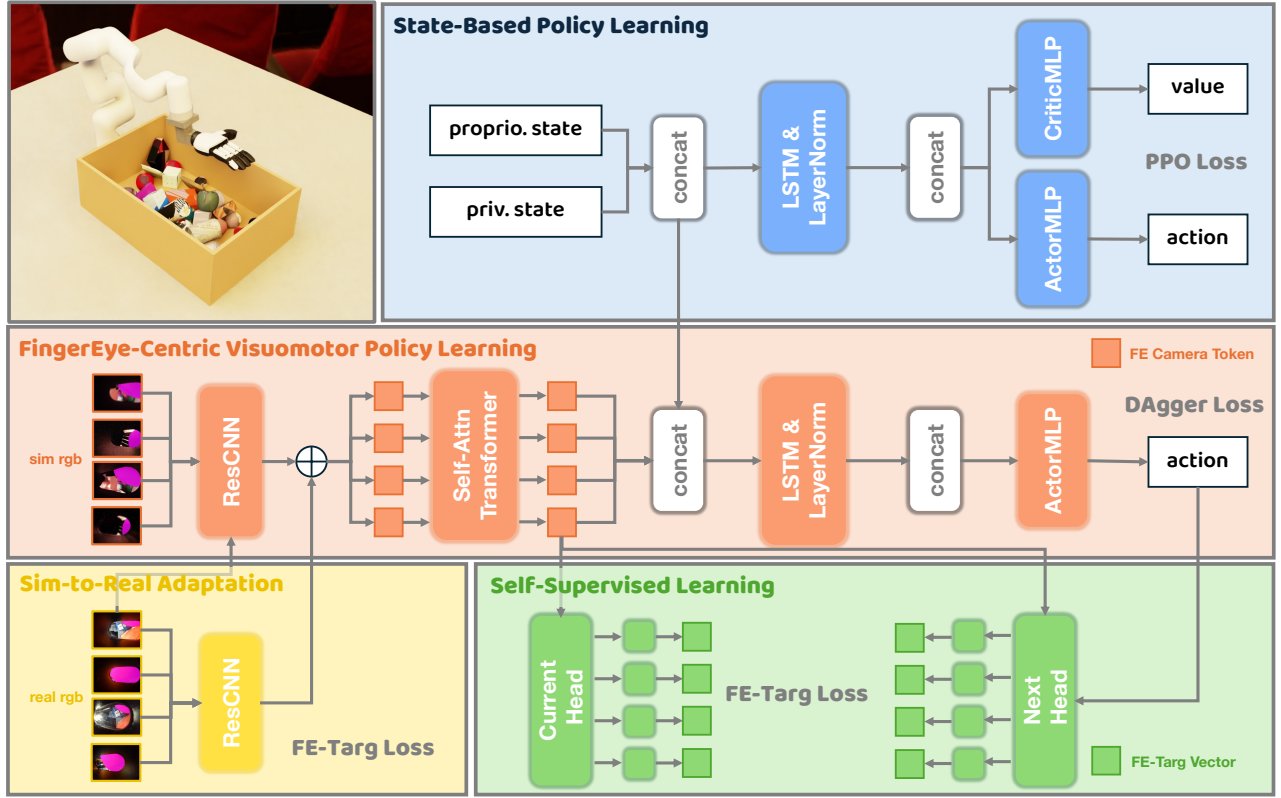


Fig. 2. Overview of DexEyeGrasp. (1) A state-based teacher learns dexterous grasping in clutter through PPO with privileged information. (2) A visual student integrates FingerEye observations and proprioceptive state, and learns action prediction through DAGger. (3) Current and action-conditioned next-step geometric prediction supervise the visual representation through the FE-Targ Loss. (4) A residual image encoder is trained on real FingerEye sequences using the same geometric objectives, while the original network remains frozen.

positions, and the nearest target–distractor planar center distance; normalized hand–target contact-force intensities; and a binary indicator of ongoing clutter settling.

C. Action Space

The policy outputs a normalized action $a_t = [a_t^{\text{trans}}, a_t^{\text{rot}}, a_t^{\text{hand}}]$, where a_t^{trans} and a_t^{rot} command end-effector translation and rotation, respectively, and a_t^{hand} controls the hand joints. Actions are scaled, saturated, and smoothed with an exponential moving average. The Cartesian command is mapped to arm-joint targets through Jacobian-based differential inverse kinematics, while the hand command is integrated into joint targets and clipped to joint limits.

D. Partial Observation

The visual student has access only to the proprioceptive state $\mathbf{x}_t^{\text{prop}}$ and visual observations. FingerEye observations consist of RGB images, while observations from the overview camera can optionally include RGB images or depth information; the latter can also be converted into a point cloud using the camera calibration. We use SAM2 to provide target identity in the visual observations, enabling the student to distinguish the designated target from distractors.

IV. METHOD

As illustrated in Fig. 2, we distill a state-based teacher into a FingerEye-centric visuomotor student, using self-supervised geometric objectives to support policy learning in simulation and residual visual adaptation to the real world.

A. State-Based Policy Learning with Privileged Information

We first train a state-based teacher in simulation using proximal policy optimization (PPO) [23]. With access to the proprioceptive and privileged states defined above, the teacher learns dexterous grasping behaviors that subsequently serve as supervision for the visual student.

1) *State-Based Policy Architecture*: The teacher uses a recurrent actor–critic architecture. An LSTM encodes the normalized state $\mathbf{s}_t$, and its output is concatenated with the current state and passed to separate actor and critic MLPs. The actor outputs an 18-dimensional Gaussian action distribution for joint arm–hand control, while the critic estimates the expected discounted return.

2) *Task Reward Design*: We group the task reward into reaching, lifting, and goal-reaching terms:

$$r_t = r_t^{\text{reach}} + r_t^{\text{lift}} + r_t^{\text{goal}}. \quad (1)$$

Each term retains the weights of its constituent rewards. The reaching term r_t^{reach} rewards reductions in the distance from a hand reference point to the target. The lifting term r_t^{lift}

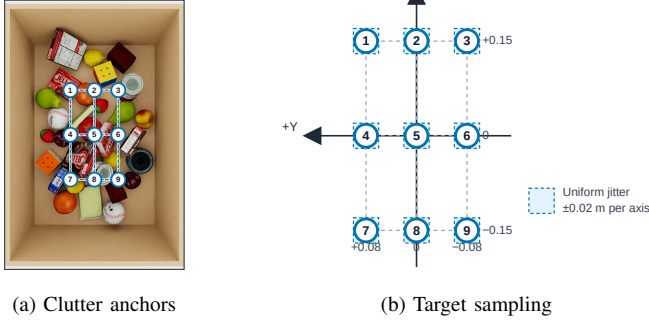


Fig. 3. The 3×3 grid curriculum. (a) Nine target-position anchors in the cluttered workspace. (b) Anchor coordinates in meters; target positions are uniformly perturbed by ± 0.02 m along each planar axis around the assigned anchor.

combines progress toward the desired lifting height with a one-time bonus for exceeding that height. The goal term r_t^{goal} combines reductions in the target-goal distance, a per-step bonus for remaining within the goal threshold, and a success bonus after H_g consecutive valid steps, as defined in Section III-A. All progress rewards are measured relative to the best previously achieved error within the episode. Lifting and goal rewards require valid target contact and hand contact forces below a prescribed limit, and all rewards are suppressed during initial clutter settling.

3) *Clutter Curriculum and Physical Randomization:* We use a 3×3 grid curriculum to ease teacher training. We first train multiple teachers with target positions perturbed around specific grid anchors across the workspace (Fig. 3). Their behaviors are distilled into a single state-based policy through multi-teacher DAgger, followed by PPO training in fully randomized clutter layouts. We then fine-tune the unified teacher with episodic physical randomization of object masses, friction coefficients, and restitution, together with the stiffness and damping gains of the arm and hand controllers. These parameters are resampled at each episode reset to encourage robustness to physical variations.

B. FingerEye-Centric Visuomotor Policy Learning

The visual student learns to reproduce the teacher’s behavior from proprioceptive signals and FingerEye observations. Our design integrates dexterous FingerEye viewpoints and incorporates self-supervised learning objectives to facilitate policy learning and improve sim-to-real transfer.

1) *Target Identification:* We encode target identity with a target-mask overlay, implemented by alpha blending a fixed color into each RGB image within the target mask while leaving the surrounding appearance unchanged. During training, we use ground-truth target masks from simulation. Deployment details are provided in Section V-A.4.

2) *Visuomotor Policy Architecture:* We encode and integrate the target-conditioned FingerEye images as

$$\begin{aligned} \mathbf{f}_{t,i} &= E(I_{t,i}), \\ (\mathbf{z}_{t,i})_{i=1}^C &= \text{Attn}((\mathbf{f}_{t,i} + \mathbf{e}_i)_{i=1}^C), \end{aligned} \quad (2)$$



Fig. 4. Examples of visual randomization in simulation. Each environment shows the scene view on the left, raw FingerEye RGB images at the upper right, and policy inputs at the lower right, with target-mask overlays and image augmentations such as blur and mosaic effects.

where $I_{t,i}$ is the image from camera i at time t , and C is the number of FingerEye cameras. The shared encoder E uses a residual CNN followed by spatial flattening and projection to produce the camera token $\mathbf{f}_{t,i}$. The learned embedding $\mathbf{e}_i$ identifies the camera, and Attn applies multi-head self-attention across views. The resulting tokens $\mathbf{z}_{t,i}$ incorporate complementary evidence from other cameras and are concatenated and projected into a fused visual feature.

This feature is concatenated with the normalized proprioceptive state $\mathbf{x}_t^{\text{prop}}$ and processed by an LSTM. An action MLP receives both the recurrent output and the current fused input, producing the 18-dimensional arm-hand command. The current input provides immediate visual feedback, while recurrence retains temporal context when the target leaves or reappears in individual views.

3) *Teacher-Student Distillation:* We train the student with online DAgger, using teacher actions as supervision at visited states. We minimize $L_{\text{imit}} = \|\mathbf{a}_t^T - \hat{\mathbf{a}}_t\|_2^2$, where $\hat{\mathbf{a}}_t$ is the student prediction, $\mathbf{a}_t^T$ is the teacher action clipped to the controller’s bounds. During training, the loss is averaged over valid samples and action dimensions.

4) *Self-Supervised Learning Objectives:* We introduce the FE-Targ Loss to encode target-relative geometry and its action-conditioned evolution through current and next-step prediction:

$$L_{\text{FE-Targ}} = \lambda_{\text{cur}} L_{\text{cur}} + \lambda_{\text{next}} L_{\text{next}}, \quad (3)$$

where λ_{cur} and λ_{next} weight the current-state and next-step prediction losses, respectively.

Let $\mathbf{y}_{t,i} = [\mathbf{u}_{t,i}, d_{t,i}] \in \mathbb{R}^4$ combine the unit camera-to-target-center direction $\mathbf{u}_{t,i}$ and distance $d_{t,i}$, normalized by a fixed maximum and clipped to $[0, 1]$. Two camera-specific heads predict this vector from the attended token $\mathbf{z}_{t,i}$ in Eq. 2:

$$\begin{aligned} \hat{\mathbf{y}}_{t,i} &= g_{\text{cur}}^i(\mathbf{z}_{t,i}), \\ \hat{\mathbf{y}}_{t+1,i} &= g_{\text{next}}^i(\mathbf{z}_{t,i}, \mathbf{a}_t). \end{aligned} \quad (4)$$

Hats denote predictions. The head g_{cur}^i predicts current geometry, while g_{next}^i additionally takes the executed action $\mathbf{a}_t$ to predict geometry in the next-step camera frame.

We define a shared geometric loss

$$L^g(\mathbf{y}, \hat{\mathbf{y}}) = 1 - \mathbf{u}^T \hat{\mathbf{u}} + \text{SmoothL1}(d - \hat{d}), \quad (5)$$

which combines cosine distance for direction and Smooth-L1 loss for distance. The current and next-step losses are then

$$\begin{aligned} L_{\text{cur}} &= L^g(\mathbf{y}_{t,i}, \hat{\mathbf{y}}_{t,i}), \\ L_{\text{next}} &= L^g(\mathbf{y}_{t+1,i}, \hat{\mathbf{y}}_{t+1,i}). \end{aligned} \quad (6)$$

Targets are obtained from simulation. Losses are averaged over valid samples and cameras with a visible target in the current image; next-step supervision also excludes episode boundaries. The joint training loss is $L = L_{\text{imit}} + L_{\text{FE-Targ}}$. The geometric prediction heads and simulation-derived targets are used only during training.

5) *Visual Randomization*: We randomize object appearance, backgrounds, and lighting (Fig. 4), and further vary image brightness, contrast, hue, saturation, noise, and blur, together with camera extrinsic perturbations. These variations expose the student to changes in scene appearance, camera response, and mounting calibration. We also retain the physical randomization used for the teacher, so action supervision is obtained under matching randomized dynamics. All randomization is applied during training; deployment uses the same visual policy without privileged simulation information.

6) *Sim-to-Real Adaptation*: We adapt the student on offline real FingerEye data (Section V-A.5) by adding a residual image encoder R_ϕ before cross-view attention:

$$\begin{aligned} \tilde{\mathbf{f}}_{t,i} &= E(I_{t,i}) + R_\phi(I_{t,i}), \\ (\tilde{\mathbf{z}}_{t,i})_{i=1}^C &= \text{Attn}((\tilde{\mathbf{f}}_{t,i} + \mathbf{e}_i)_{i=1}^C). \end{aligned} \quad (7)$$

We feed $\tilde{\mathbf{z}}_{t,i}$ to the frozen current and next-step heads and optimize only the residual parameters ϕ with $L_{\text{FE-Targ}}$. All original network parameters remain fixed; the learned residual branch is retained at deployment.

V. EXPERIMENTS

We conduct experiments in both simulation and the real world to answer the following four questions:

- **Q1**: How does DexEyeGrasp compare with existing methods for dexterous grasping in clutter?
- **Q2**: How do FingerEye observations affect grasping performance compared with overview observations and their combinations?
- **Q3**: How well does DexEyeGrasp transfer to real-world cluttered environments?
- **Q4**: Do the proposed FE-Targ Learning Objectives improve sim-to-real transfer?

A. Experimental Setup

1) *Experimental Platform and Environments*: We use an xArm7-XHand platform with FingerEye cameras and simulate it in Isaac Lab [24]. Combining 15 YCB shapes with over 300 textures yields over 4,500 object instances. *Clutter on a table* uses a $0.60 \times 0.60\text{m}$ surface with a 0.05m retaining rim; *clutter in a box* uses a $0.60 \times 0.40\text{m}$ surface with 0.20m walls. Each contains 15 or 30 objects (C15/C30), including one target.



Fig. 5. Real-world experiments. (a) The xArm7-XHand platform with FingerEye cameras and side-view sensing around a cluttered bin. (b) Target objects used for real-world grasping evaluation.

TABLE I

SIMULATION SUCCESS RATES (%) UNDER DIFFERENT CLUTTER DENSITIES AND ENVIRONMENT CONFIGURATIONS. C15/C30 DENOTE 15/30 OBJECTS. THE STATE-BASED TEACHER IS TRAINED ONLY IN CLUTTER IN A BOX WITH C30 AND DIRECTLY EVALUATED IN ALL OTHER SETTINGS WITHOUT FINE-TUNING. PARENTHESES SHOW EACH VISUAL METHOD’S CHANGE FROM THE TEACHER IN PERCENTAGE POINTS.

Method	On a table		In a box	
	C15	C30	C15	C30
Teacher	88	92	85	87
O-PC-Diff	9 (−79)	13 (−79)	8 (−77)	8 (−79)
O-PC-Trans	0 (−88)	0 (−92)	0 (−85)	0 (−87)
Ours	77 (−11)	78 (−14)	72 (−13)	76 (−11)

TABLE II

ABLATION OF DIFFERENT VISUAL OBSERVATION SOURCES UNDER THE C30 SETTINGS. O-PC, O-RGB, AND FE-RGB DENOTE THE OVERVIEW POINT CLOUD, OVERVIEW RGB OBSERVATIONS, AND FINGEREYE RGB OBSERVATIONS, RESPECTIVELY. ENTRIES ARE SUCCESS RATES (%); PARENTHESES SHOW REAL-MINUS-SIMULATION DIFFERENCES IN PERCENTAGE POINTS.

Visual Sources	On a table		In a box	
	Sim	Real	Sim	Real
O-PC	36	0 (−36)	34	0 (−34)
O-PC + FE-RGB	45	30 (−15)	44	20 (−24)
O-RGB	39	0 (−39)	52	0 (−52)
O-RGB + FE-RGB	75	60 (−15)	71	50 (−21)
FE-RGB (Ours)	78	80 (+2)	76	80 (+4)

2) *Baselines*: Both baselines follow teacher-student distillation. For a matched comparison, all students use our state-based teacher and online DAgger. Baselines receive overview point clouds (O-PC); ours receives FingerEye RGB. All use proprioception. The teacher serves only as a privileged-information reference.

- **O-PC-Trans**, adapted from DexSinGrasp [1], uses a

TABLE III

SIMULATION AND REAL-WORLD SUCCESS RATES (%) UNDER DIFFERENT CLUTTER DENSITIES AND ENVIRONMENT CONFIGURATIONS. PARENTHESES SHOW REAL-MINUS-SIMULATION DIFFERENCES IN PERCENTAGE POINTS.

Method	Clutter on a table				Clutter in a box			
	C15		C30		C15		C30	
	sim	real	sim	real	sim	real	sim	real
O-PC-Diff	9	0 (−9)	13	0 (−13)	8	0 (−8)	8	0 (−8)
O-PC-Trans	0	0 (0)	0	0 (0)	0	0 (0)	0	0 (0)
Ours	77	90 (+13)	78	80 (+2)	72	90 (+18)	76	80 (+4)

TABLE IV

ABLATION OF THE FE-TARG LOSS AND FE-TARG ADAPTATION UNDER THE C30 SETTINGS. ENTRIES ARE SUCCESS RATES (%); PARENTHESES SHOW REAL-MINUS-SIMULATION DIFFERENCES IN PERCENTAGE POINTS.

DASHES INDICATE SIMULATION RESULTS IDENTICAL TO OURS: THE RESIDUAL IMAGE ENCODER LEARNED DURING ADAPTATION IS USED ONLY FOR REAL IMAGES.

Method	On a table		In a box	
	Sim	Real	Sim	Real
Ours w/o FE-Targ Loss	63	70 (+7)	66	70 (+4)
Ours w/o FE-Targ Adapt.	—	80 (+2)	—	60 (−16)
Ours	78	80 (+2)	76	80 (+4)

frozen pretrained point encoder [25] and a Transformer action predictor. To accommodate randomized target placement, we provide target-centered XYZ point-cloud crops using overview depth and target masks (ground truth in simulation, SAM2 at deployment), and replace offline behavior cloning with DAGger.

- **O-PC-Diff**, based on ClutterDexGrasp [2], uses XYZ point clouds with a target-identity channel. Without an available official implementation, we reproduce its architecture using DP3 [12] and replace offline distillation with DAGger, retaining the diffusion objective.

All baseline results refer to these adapted versions.

3) *Training and Evaluation Protocol*: Teacher and student train at 15Hz in simulation. Success requires meeting the grasping and goal-reaching criterion in Sec. III-A within the episode limit. We report mean success rates over 100 simulation episodes per seed across three seeds (300 total), and success percentages over 10 real-world trials per method. Bold marks the best visual-policy results (including ties); blue marks ours.

4) *Real-World Deployment*: A side-view Intel RealSense D435 complements FingerEye sensing (Fig. 5(a)). Real clutter contains over 50 distinct objects, including over 20 targets (Fig. 5(b)). SAM2 [9] tracks targets from manual first-frame bounding boxes, retaining only non-empty-mask frames in memory. Initially target-absent views use their current frame with valid histories from other views until a mask appears, then use their own valid histories. Empty-mask frames remain unmodified policy inputs and do not update memory. Multi-view SAM2 processing reduces control from

the simulated 15Hz to approximately 10Hz on hardware. Figure 6 shows successful and failed deployment trajectories alongside FingerEye observations and target-mask overlays.

5) *Real-World Data Collection*: We collect approximately 50 teleoperated approach-and-grasp trajectories, recording FingerEye RGB, proprioception, and overview RGB-D. Successful grasps are unnecessary: adaptation supervises geometry, not action imitation. Scanned textured object meshes, overview RGB-D, and a manual first-frame bounding box enable FoundationPose [26] to estimate target poses. Forward kinematics and FingerEye extrinsics give camera poses; expressing target centers in camera coordinates yields direction–distance labels $\mathbf{y}_{t,i}$, normalized as in simulation. Reconstructed actions are normalized to one 15Hz control step for next-step prediction. Images are resized without cropping. Supervision uses non-empty-mask views; next-step prediction additionally excludes invalid temporal transitions.

B. Simulation Performance

To answer **Q1**, Table I compares students distilled from the same teacher. DexEyeGrasp achieves 72–78% success across configurations, versus 8–13% for O-PC-Diff and 0% for O-PC-Trans, remaining within 11–14 percentage points of the privileged teacher. In C30, our policy achieves 78% on a table and 76% in a box.

The low baseline scores may partly reflect adaptation mismatches. O-PC-Trans has no explicit target-identity input and assumes the target is centered in the scene point cloud; target-centered cropping approximates this assumption but may leave target ambiguity in dense clutter. Both baseline architectures were originally used for offline imitation and may be less suited to our shared online DAGger protocol. These results thus characterize the adapted baselines rather than their original settings. To reduce architectural confounds, the following ablation holds the policy backbone fixed while varying visual sources and their encoders.

C. Visual Observation Source Ablation

To answer **Q2**, we compare O-PC, O-RGB, their combinations with FE-RGB, and FE-RGB alone in both C30 settings and domains. All variants share our policy backbone and training settings, using a DP3 encoder for point clouds and ResCNN for RGB; visual embeddings are concatenated with proprioceptive state.



Fig. 6. Real-world grasping trajectories: successful trials with an orange ball and a pink paper cup, and a failed trial with a red object. Each example shows the target object on the left and, on the right, overview images, middle-finger FingerEye RGB images, and target-mask overlays from top to bottom. Time progresses left to right; overview and FingerEye columns are not synchronized.

Table II shows that FE-RGB alone achieves 78%/76% in simulation and 80%/80% on hardware for table/box clutter. O-PC achieves 36%/34% in simulation and no real-world successes. Adding FE-RGB raises its simulation success to 45%/44%, gains of 9/10 percentage points, and its real-world success to 30%/20%. O-RGB + FE-RGB achieves 75%/71% in simulation and 60%/50% on hardware. Both combinations improve on their overview-only counterparts, but neither outperforms FE-RGB alone.

In principle, overview observations provide complementary scene information; an effective fusion mechanism could exploit it or ignore unhelpful inputs, matching or improving on FE-RGB alone. However, realizing this benefit is non-trivial: point clouds and RGB encode geometry and appearance differently, and concatenating their embeddings does not ensure useful cross-modal integration. With O-RGB, the deficit relative to FE-RGB alone grows from 3/5 percentage points in simulation to 20/30 points on hardware. This is consistent with an additional sim-to-real gap from overview RGB appearance, although the ablation does not isolate its cause. Thus, additional sensing does not automatically

improve the learned policy; fusion and visual transfer remain important. FE-RGB and proprioception are sufficient policy inputs in the evaluated settings.

D. Sim-to-Real Transfer

To answer **Q3**, we evaluate the sim-to-real transfer of DexEyeGrasp across all clutter densities and environment configurations. We compare DexEyeGrasp with O-PC-Diff and O-PC-Trans, and report both simulation and real-world success rates.

Table III reports real-world success rates of 90% and 80% on a table and 90% and 80% in a box for C15 and C30, respectively. Neither adapted baseline succeeds in the real-world trials. All four real-world results exceed their simulation counterparts, with success rates of 80–90% on hardware. Overall, the policy transfers to real-world clutter, although the limited number of trials calls for further experiments to validate these results.

E. FE-Targ Learning Objectives Ablation

To answer **Q4**, we investigate the contributions of the FE-Targ Loss during simulation training and FE-Targ adaptation

during sim-to-real transfer. The FE-Targ Loss encourages the visual representation to capture target-relevant information from dynamic FingerEye observations.

We compare DexEyeGrasp (Ours) with variants trained without the FE-Targ Loss or evaluated without FE-Targ adaptation. This ablation is conducted under the two C30 settings, where the high object density and wall constraints create the largest perception and manipulation challenges.

Table IV shows a consistent simulation benefit from the FE-Targ Loss: success increases from 63% to 78% on a table and from 66% to 76% in a box, gains of 15 and 10 percentage points. These gains support geometric supervision alongside action imitation. Notably, FE-Targ adaptation leaves performance unchanged at 80% in the easier setting on a table, but improves success by 20 percentage points, from 60% to 80%, in the more challenging setting in a box. This result suggests that adaptation is particularly beneficial when severe occlusion and constrained object motion exacerbate the visual domain gap. Results from additional trials will be included in the supplementary material.

F. Failure Modes and Limitations

The failed trajectory in Fig. 6 may reflect the combined effects of physical blockage and reduced control frequency. More generally, we observe three limitations. First, prolonged target invisibility or tracking drift to a distractor deprives the policy of reliable target cues, despite retaining valid SAM2 frames. Second, distractors or walls can block hand-object motion and prevent grasp completion. Third, multi-view segmentation and tracking reduce control from 15Hz in simulation to approximately 10Hz on hardware, potentially delaying corrective actions during contact. The illustrated failure does not isolate these factors individually.

VI. CONCLUSION

We presented DexEyeGrasp, a FingerEye-centric framework for dexterous grasping in dense clutter. The framework distills a state-based teacher into a visual student that integrates multiple FingerEye observations to control both finger and wrist motion. Self-supervised current and next-step geometric prediction encourages target-relative spatial awareness, while the same objectives support residual visual adaptation using real teleoperated trajectories without direct action imitation. Future work will investigate more reliable target tracking, recovery from blocked interactions, and faster multi-view perception for closed-loop control.

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